Responsibility and verification of AI-generated code in online discussions

Group 36 COMP3020 Social Web Analytics Hacker News discussion corpus

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Overall question. How are responsibility and verification of AI-generated code discussed online, and how are these discussions structured through reply interactions?

5 selected threads **313 raw comments** **278 retained comments** **186 authors**

Data and methods

The group reports collection with R on 28 Tháng Chín 2026. Public Hacker News comments: 30 dead and 5 deleted records removed. IDs and parent links retained; three threads contribute 94.6% of comments.

Reply outcome: at least one direct child comment also retained in the clean corpus.

RQ1 What is discussed?

What responsibilities and verification practices are discussed in relation to AI-generated code?

Word counts, six non-exclusive lexical indicators and contextual reading describe the discussion.

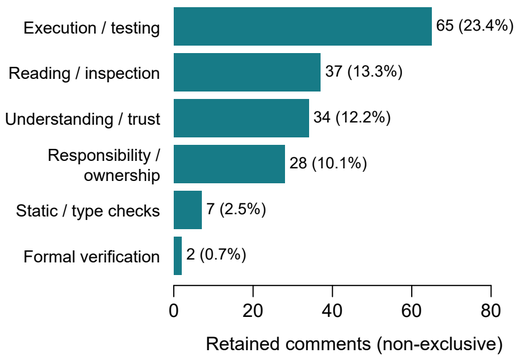


Figure 1. Candidate mentions among all 278 comments.

Most frequent words: not (335), code (286), ai (166). Frequency does not establish endorsement.

Context matters. Ownership and review appear together (43861845, 43864090); another comment questions the cost of extra responsibility to reviewers (43867077).

Checking appears alongside ownership and trust. Context remains essential.

Foundation. Barke et al. (2023) motivate the practice categories. O'Brien (2025) cautions against equating reported verification with successful practice.

RQ2 Verification and replies

Is mentioning a specific verification practice associated with receiving a direct reply?

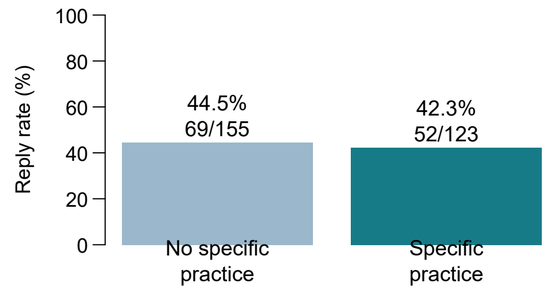


Figure 2. Comments with a retained direct reply.

Difference (specific minus no specific): **-2.24 percentage points**.

Yates-corrected $\chi^2(1) = 0.064, p = 0.8008$. Smallest expected count: 53.5.

First per author: +0.80 points. AI rule (excluding 5 ambiguous): +1.08. Small contrasts change direction; dependence remains.

RQ3 Content clusters

What content clusters emerge, and which aspects of responsibility or verification characterise them?

Log TF-IDF, L2 normalisation, cosine distance and average-link hierarchical clustering of 264 eligible comments.

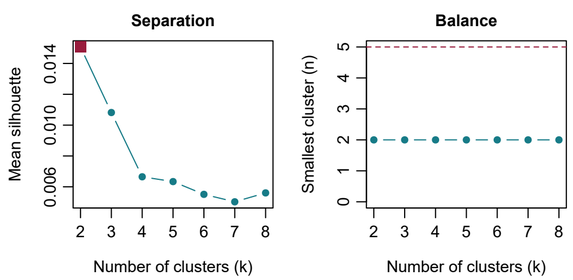


Figure 3. Separation and size diagnostics for $k = 2$–8.

Selected $k = 2$ as a diagnostic fallback; mean silhouette **0.015**. Cluster sizes: **262 and 2**.

The tiny pair uses 'hand' in different senses. Three representation checks also fail to support clear themes.

RQ4 Who and what gets replies?

How are replies structured, and how do they connect to content clusters?

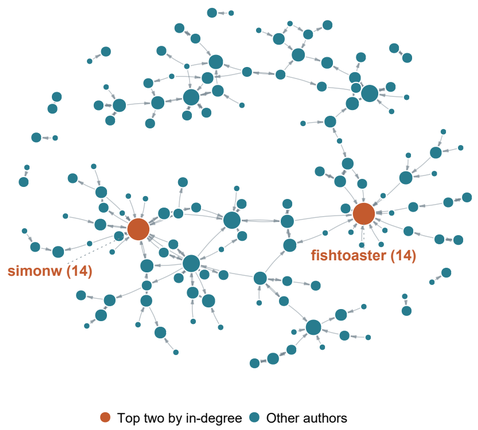


Figure 4. Full reply network. Larger nodes receive replies from more distinct authors; orange highlights the two leading targets.

144 authors **193 directed pairs**

198 matched reply events. The largest weak component contains 128 authors (88.9%). 42 corpus authors have no retained between-author edge.

Two structural measures

Author	In-degree	Betweenness
simonw	14	0.0191
fishtoaster	14	0.0030

Table 1. Distinct replying authors and directed, unweighted, normalised betweenness in the full graph.

Content bridge. Comments receiving replies by lexical group: C1: 118/262 (45.0%); C2: 0/2 (0.0%). Weak and unequal clusters prevent a reliable thematic comparison.

Replies to one verification proposal include both objections and qualified support. Prominence does not imply approval.

Main findings

Content. The selected discussions connect checking code with ownership, review and trust.

Response. The reply-rate contrast changes direction with author weighting and an explicit AI-assisted coding rule. A stable directional relationship is not established.

Structure. Most network participants share one large weak component, while lexical clustering separates comments poorly.

The strongest evidence is descriptive: what these participants discuss and who replies to whom. Stronger claims require broader data and better measurement.

Limits and next steps

Five selected, uneven threads cannot represent all programmers or workplace behaviour.

Thread and author dependence, ambiguous text, deleted or missing parents, and short comments constrain inference.

The original Excel rule remains uninspected. A separate AI-assisted reading flags 14 differences and 5 ambiguous cases; it is not human validation.

Broader thread sampling, documented annotation and contextual checks should precede stronger substantive conclusions.

Reproducibility and sources

Frozen input data, keyed joins, R scripts, parameters and complete results accompany the report.

GitHub repository

https://github.com/ledinhtrung05-coder/COMP3020-Group36

Barke, James & Polikarpova (2023). *Grounded Copilot*. DOI: 10.1145/3586030.

O'Brien (2025). *How Scientists Use Large Language Models to Program*. DOI: 10.1145/3706598.3713668.

Grimmer & Stewart (2013). *Text as Data*. DOI: 10.1093/pan/mps028.

Data: Hacker News API. Full references and methods appear in the report.

REVIEW BUILD: complete member details and group review; collection date is group-confirmed and annotation limitations are disclosed.